

Mandatory Assignment 2

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April 11, 2024

1 Exercise

1.1 Exercise 1.

The goal here is to construct a clean dataset for cross-sectional stock return prediction. We keep observations from January 2000 onward, select a limited set of influential predictors, document missing-value handling, and prepare a feature matrix with characteristic - macro interactions and industry dummies.

1.1.1 1a: Variable selection and sample restriction

First we read in the monthly stock characteristics data and restricting the sample to observations dated on or after January 1, 2000.

We select ten stock characteristics and four macro variables capturing key GDX themes and macroeconomic conditions, including valuation, interest rates, and credit risk.

The selected characteristics are economically interpretable and widely used in asset pricing. In particular, mom12m, mvel1, bm, and roaq capture momentum, size, value, and profitability. These are complemented by variables reflecting valuation, trading activity, risk, investment, and accounting quality. The macro variables summarize market valuation, interest rates, and credit conditions, which are standard predictors of returns.

1.1.2 1b: Summary Statistics and Missing Values

Table 1 reports summary statistics for the selected predictors.

	mean	std	min	25%	50%	75%	max
characteristic_mom12m	-0.007	0.590	-0.997	-0.526	-0.026	0.519	0.992
characteristic_mom1m	-0.004	0.604	-0.996	-0.550	-0.009	0.544	0.995
characteristic_mvel1	0.026	0.588	-1.000	-0.496	0.044	0.544	1.000
characteristic_bm	-0.065	0.531	-0.990	-0.512	-0.067	0.373	0.990

	mean	std	min	25%	50%	75%	max
characteristic_roaq	-0.014	0.560	-0.991	-0.477	-0.023	0.457	0.990
characteristic_ep	-0.011	0.558	-0.990	-0.486	-0.004	0.454	0.990
characteristic_turn	0.036	0.576	-1.000	-0.453	0.074	0.532	0.994
characteristic_beta	0.076	0.548	-0.993	-0.370	0.102	0.540	0.993
characteristic_invest	-0.006	0.533	-0.990	-0.427	-0.000	0.413	0.990
characteristic_acc	-0.008	0.545	-0.990	-0.445	-0.023	0.437	0.990
macro_dp	-4.022	0.208	-4.524	-4.107	-4.001	-3.904	-3.281
macro_ep	-3.177	0.382	-4.836	-3.317	-3.113	-2.917	-2.566
macro_tbl	0.018	0.019	0.000	0.001	0.011	0.028	0.062
macro_dfy	0.010	0.004	0.005	0.008	0.009	0.012	0.034

The selected stock characteristics are approx centered around zero and bounded close to the $[-1,1]$ interval, which is consistent with the scaling used in the provided dataset. The macro variables display much less cross-sectional variation because they are common state variables observed at the monthly level.

To document data quality, we first count missing values before imputing.

	n_missing
characteristic_mom12m	0
characteristic_mom1m	0
characteristic_mvell	0
characteristic_bm	0
characteristic_roaq	0
characteristic_ep	0
characteristic_turn	0
characteristic_beta	0
characteristic_invest	0
characteristic_acc	0
macro_dp	0
macro_ep	0
macro_tbl	0
macro_dfy	0

Missing stock characteristics are imputed using the monthly cross-sectional median, preserving contemporaneous information. Macroeconomic variables are forward- and backward-filled. Observations with missing excess returns or lagged market capitalization are dropped, as they are required for the analysis.

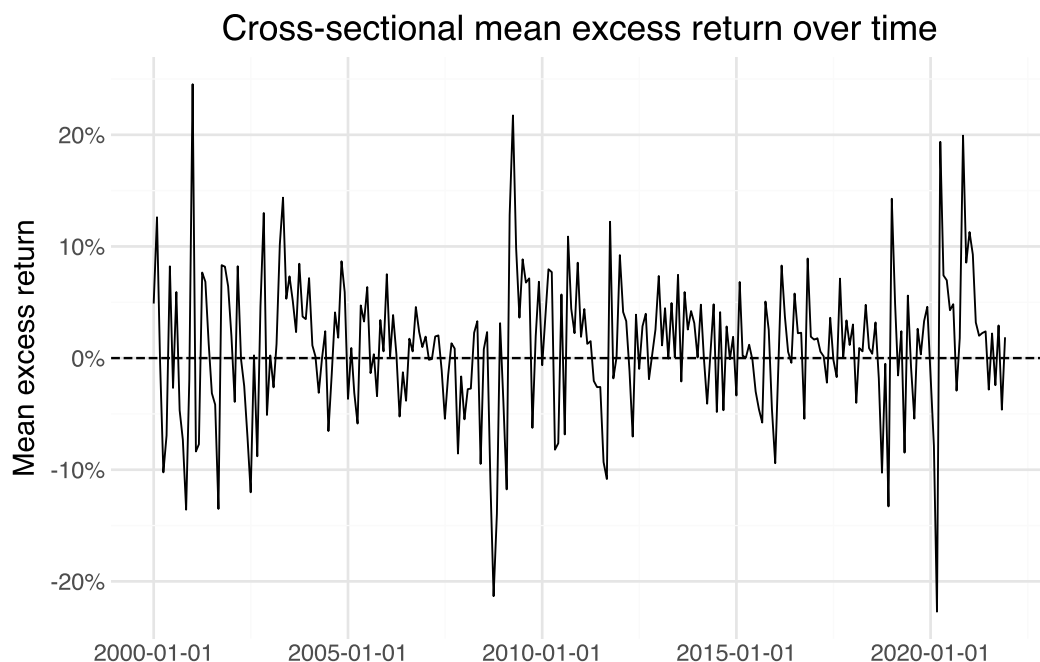
	n_missing
characteristic_mom12m	0
characteristic_mom1m	0
characteristic_mvell	0

	n_missing
characteristic_bm	0
characteristic_roaq	0
characteristic_ep	0
characteristic_turn	0
characteristic_beta	0
characteristic_invest	0
characteristic_acc	0
macro_dp	0
macro_ep	0
macro_tbl	0
macro_dfy	0
ret_excess	0
mktcap_lag	0

Because the assignment allows the use of sic2, we recode the detailed SIC groups into the 10 major SIC divisions. This keeps the dimensionality manageable while retaining broad industry information.

	sic_division	n_obs
0	Manufacturing	442708
1	Finance	224201
2	Services	194522
3	Transportation	86802
4	Retail	64355
5	Mining	42998
6	Wholesale	31682
7	Construction	12262
8	Public Admin	9530
9	Agriculture	2545

As a simple illustration of the target variable, Figure 1 plots the cross-sectional mean excess return by month.



The sample remains large and suitable for machine learning. Missing values are handled conservatively, with firm-level predictors imputed cross-sectionally and macro variables filled over time.

1.1.3 1c: Feature matrix construction

The assignment asks for a feature matrix consisting of stock characteristics and all interactions between stock characteristics and macroeconomic variables. With 10 stock characteristics and 4 macro predictors, this yields $10+10\times 4=50$ continuous predictors. We then add industry dummies based on the recoded SIC divisions.

	object	shape
0	X	(1111605, 59)
1	y	(1111605,)
2	meta	(1111605, 4)

	feature_group	count
0	stock characteristics	10
1	interaction terms	40
2	industry dummies	9
3	total features	59

The final feature matrix contains the 10 selected stock characteristics, 40 interaction terms between stock characteristics and macroeconomic variables, and industry dummies for the broad SIC divisions.

1.2 Exercise 2

1.2.1 Exercise 2a: Model descriptions

Elastic Net is a penalized linear regression model that combines the L_1 penalty from the Lasso and the L_2 penalty from Ridge regression. It minimizes the following objective function:

$$\min_{\beta} \frac{1}{N} \sum_{i=1}^N (y_i - x_i' \beta)^2 + \alpha (\lambda \|\beta\|_1 + (1 - \lambda) \|\beta\|_2^2)$$

where the first term is the mean squared prediction error and the second term is a penalty on coefficient size. This helps control overfitting in high-dimensional settings with correlated predictors. This is particularly relevant in our setting, as the feature matrix includes stock characteristics, interaction terms, and industry dummies.

The key hyperparameters are the overall penalty strength (**alpha**) and the mixing parameter (**l1_ratio**), where **l1_ratio** = 1 corresponds to the Lasso and **l1_ratio** = 0 corresponds to Ridge.

Neural Network (MLP) approximates $g(z_{i,t})$ as a composition of linear transformations and nonlinear activation functions across multiple hidden layers. The model is trained by minimizing the mean squared prediction error between predicted and realized excess returns. Each neuron computes $x_k^l = f\left(\theta_0^k + \sum_j z_j \theta_j^k\right)$, where f is the ReLU activation function. The network is trained via backpropagation. The key hyperparameters are the number and size of hidden layers (**hidden_layer_sizes**), the L_2 regularization strength (**alpha**), and the maximum number of iterations (**max_iter**).

The two models provide a comparison between a structured, regularized linear benchmark and a more flexible nonlinear model.

1.2.2 Exercise 2b: Time-respecting data split

Train: 2000-01-01 to 2013-02-01 (748,902 obs)
Validation: 2013-03-01 to 2017-07-01 (194,460 obs)
Test: 2017-08-01 to 2021-12-01 (168,243 obs)
Number of train months: 158
Number of validation months: 53
Number of test months: 53

The dataset is split chronologically to avoid look-ahead bias, ensuring that training observations precede validation observations, which in turn precede test observations. The most recent 20% of months are reserved as the test set, providing a truly out-of-sample evaluation period.

The remaining 80% is split into 75% training and 25% validation, yielding an approximate 60/20/20 split. This reflects a trade-off between reliable parameter estimation and effective model selection.

The split is done at the monthly level to preserve the panel structure. Continuous predictors are standardized using the training data only, and the same transformation is applied to validation and test sets to avoid data leakage.

1.2.3 Exercise 2c: Hyperparameter tuning

Elastic Net tuning results:

alpha	l1_ratio	train_rmse	val_rmse
0.100	0.2	0.199689	0.157523
0.100	0.8	0.199689	0.157523
0.001	0.8	0.199399	0.157715
0.001	0.2	0.199213	0.157819

Best Elastic Net params: {'alpha': 0.1, 'l1_ratio': 0.2}

Neural Network tuning results:

hidden_layers	alpha	train_rmse	val_rmse
(32, 16)	0.001	0.192807	0.158464
(32, 16)	0.010	0.196407	0.158962

Best NN: layers=(32, 16), alpha=0.001

Final models fitted successfully.

Elastic Net - val RMSE: 0.157523

Neural Network - val RMSE: 0.158464

Hyperparameters are tuned using grid search, where each specification is evaluated on the validation set using RMSE, consistent with the MSPE objective in GKK. The Elastic Net grid varies both penalty strength and the balance between L_1 and L_2 regularization, while the Neural Network grid varies the regularization strength for a fixed architecture.

The results illustrate the bias-variance trade-off: lower training RMSE does not necessarily imply lower validation RMSE. Weaker regularization may lead to overfitting, while stronger regularization increases bias but can improve generalization.

The final model for each method is selected based on the lowest validation RMSE to maximize out-of-sample predictive performance.

1.2.4 Exercise 2d: Save models to disk

```
['model_neural_network.joblib']
```

To load the models for peer review, run:

```
import joblib
final_en = joblib.load("model_elastic_net.joblib")
final_nn = joblib.load("model_neural_network.joblib")
```

The final fitted models are saved to disk using the `joblib` library to ensure reproducibility and facilitate peer review. Saving the trained models allows others to load the exact same fitted objects without re-running the entire estimation procedure, which can be computationally intensive given the large dataset.

To load the models for replication or further analysis, the following code can be used:

1.3 Exercise 3

1.3.1 Exercise 3a: Generate predicted excess returns

Using the fitted models, we generate predicted excess returns for each stock-month observation in the validation and test samples. These predictions reflect expected returns based on firm characteristics, macroeconomic variables, and their interactions. The predictions are used to rank stocks in portfolio construction, where higher predicted returns correspond to higher deciles and more attractive investment opportunities.

1.3.2 Exercise 3b: Decile portfolio sorts

Stocks are sorted into decile portfolios each month based on predicted excess returns. Within each month, stocks are ranked and divided into ten groups of approximately equal size, from the lowest (decile 0) to the highest (decile 9).

To ensure stable portfolio formation, predictions are ranked before applying quantile sorting, avoiding issues with tied values. Months with very few observations are excluded.

This procedure generates a cross-sectional ranking that reflects the model's expected return signal and is used to construct long-short portfolios.

1.3.3 Exercise 3c: Long-short portfolio

Long-short portfolios constructed.
Validation period: 2013-03-01 to 2017-07-01
Test period: 2017-08-01 to 2021-12-01
Number of valid validation months (EN): 53
Number of valid test months (EN): 53

1.3.4 Exercise 3d: Performance metrics

Elastic Net - Validation
Annualized mean excess return: 0.0410 (4.10%)
Annualized Sharpe ratio: 0.6976
CAPM alpha (annualized): 0.0405 (4.05%)
CAPM alpha t-stat: 1.3369

Neural Net - Validation

Annualized mean excess return: -0.0507 (-5.07%)
Annualized Sharpe ratio: -0.4420
CAPM alpha (annualized): -0.0922 (-9.22%)
CAPM alpha t-stat: -1.6196

Elastic Net - Test

Annualized mean excess return: 0.0641 (6.41%)
Annualized Sharpe ratio: 0.7228
CAPM alpha (annualized): 0.0449 (4.49%)
CAPM alpha t-stat: 1.0329

Neural Net - Test

Annualized mean excess return: -0.0382 (-3.82%)
Annualized Sharpe ratio: -0.2336
CAPM alpha (annualized): -0.0769 (-7.69%)
CAPM alpha t-stat: -0.9641

Market benchmark - Validation

Annualized mean: 13.61%
Sharpe ratio: 1.3293

Market benchmark - Test

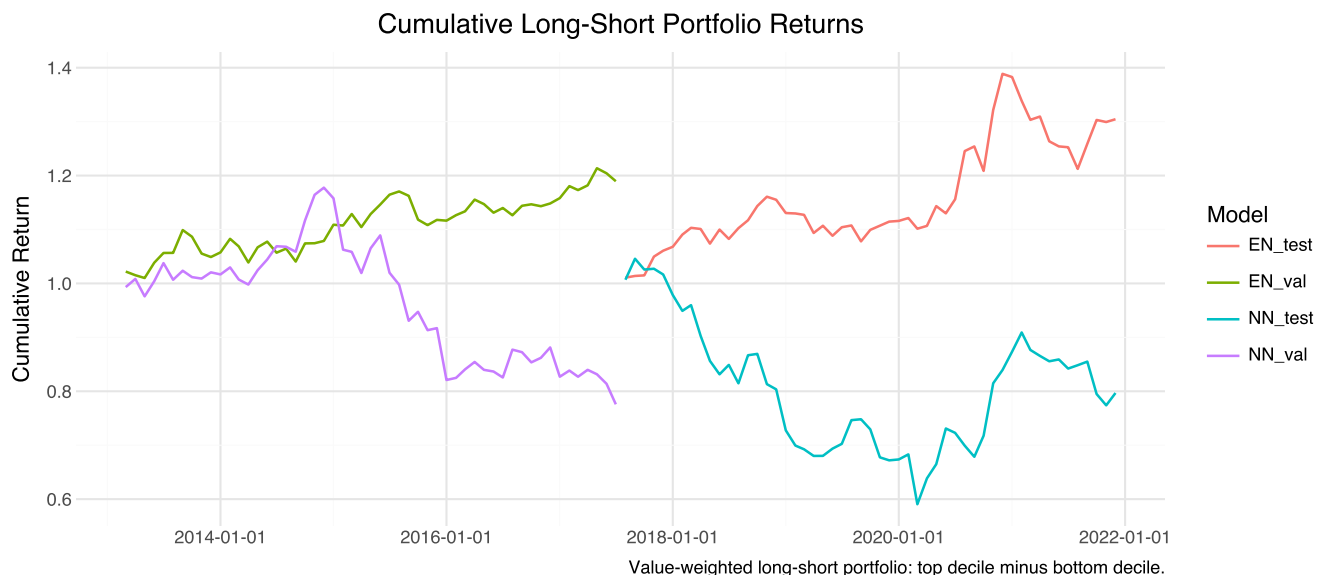
Annualized mean: 17.48%
Sharpe ratio: 1.0233

1.3.5 Exercise 3e: Portfolio turnover

Monthly turnover - Elastic Net: Val=0.007, Test=0.008

Monthly turnover - Neural Network: Val=0.341, Test=0.408

1.3.6 Exercise 3f: Model comparison



Both models are compared across Sharpe ratio, CAPM alpha, and portfolio turnover. The Elastic Net is the linear benchmark and tends to have lower turnover due to its smooth, stable coefficient structure. The Neural Network captures nonlinear interactions but may overfit to short-lived patterns, leading to higher turnover and potentially lower net-of-cost returns. If the Neural Network's alpha does not sufficiently exceed the Elastic Net's to compensate for higher turnover costs, the Elastic Net is preferable from an investor's perspective.

1.4 Exercise 4

1.4.1 Exercise 4a: Variable importance

	model	feature	sharpe_drop
0	Elastic Net	characteristic_mom12m	0.0000
1	Elastic Net	characteristic_invest_x_macro_tbl	0.0000
2	Elastic Net	characteristic_ep_x_macro_tbl	0.0000
3	Neural Network	characteristic_beta_x_macro_ep	0.3407
4	Neural Network	sic_Manufacturing	0.0572
5	Neural Network	sic_Retail	0.0368

Variable importance is measured as the drop in annualized test Sharpe ratio when one predictor is set to its mean and the long-short portfolio is rebuilt. For the Elastic Net, the top three predictors all have a Sharpe-ratio drop of 0.0000, indicating no meaningful variable importance under this metric. For the Neural Network, the most important predictors are `characteristic_beta_x_macro_ep`, `sic_Manufacturing`, and `sic_Retail`.

1.4.2 Exercise 4b: Ranking comparison

	feature	rank_en	rank_nn	rank_gap
13	characteristic_bm_x_macro_ep	1.0	59.0	58.0
16	characteristic_ep_x_macro_dfy	1.0	58.0	57.0
15	characteristic_ep	1.0	57.0	56.0
12	characteristic_bm_x_macro_dp	1.0	56.0	55.0
31	characteristic_mom1m_x_macro_dfy	1.0	55.0	54.0

The rankings differ strongly across models. The Elastic Net shows almost no sensitivity to individual predictors, while the Neural Network places importance on one interaction term and two industry dummies. This likely reflects the Neural Network's greater flexibility. However, its results should be interpreted cautiously because its test Sharpe ratio is negative.

1.4.3 Exercise 4c: Findings to financial theory

The main Neural Network signal is `characteristic_beta_x_macro_ep`, which suggests state-dependent predictability: the pricing of market beta varies with aggregate valuation conditions. The importance of `sic_Manufacturing` and `sic_Retail` indicates that industry exposure also matters. Overall, the findings point more to state-dependent and industry-specific effects than to strong standalone linear premiums.

References